



LEO New Approach Acquisition

LEO Acquisition Themes

- Procurement will occur in three phases:
 - Phase 1: Core and 2 Commercial Modules
 - Phase 2: Market Driven Power and Cooling Module
 - Phase 3: Market Driven Commercial Expansion or Free Flight Services
- Procure a NASA-owned Core Module via a supply contract with potential partnership opportunities.
 - Plan to attach to the International Space Station (ISS) or if needed, go to free flight directly.
- Develop two commercial modules via partnership arrangements.
 - These modules will be attached to the NASA-owned Core Module while attached to ISS
- NASA will continue to procure Commercial Space Transportation services (Crew and Cargo)
 - The plan will ensure transportation capabilities remain viable between the arrival of the new core station attached to ISS and the end of the ISS.

Transition to services will only occur when market demand matures and is self sustaining
Commercial Modules will be a unique partnership arrangement between NASA and the Provider

Core Module Acquisition Strategy

Core Module Supply Based Contract Acquisition

- Primary objective of the Core Module acquisition is to build, delivery, maintain and sustain the Core.
- May initially select 2 awardees for the requirements and architectural maturation phase
- Downselect to 1 after architectural maturation

Rapid High TRL Dev Incentive Industry To:

- Use High Reliability and TRL components
- Maximize use industry standards that meet intent of NASA standards.
- Tailor design approach for rapid iteration and early testing.

Partnering (Cost Offset to NASA for Core Module)

- NASA seeks industry ideas to reduce/offset cost for the build, deliver and or sustaining of the Core Module
- Access to or use of ISS, Crew Seats, use of the Core, etc.
- Value of the offsets would result in an equitable cost reduction

NASA wants to engage in new and innovative ways to leverage commercial cost offsets and partnership opportunities even on the Core Module

Potential Example for Core Module Cost Offset

- Provider bids a cost offset to the Core Module development based on NASA provided commercial opportunities
 - Provider requests an ISS seat on a future long duration mission with a split of their crew time between NASA and Commercial Tasks
 - Additionally, they request access to an amount of cargo upmass and EXPRESS rack space for their own commercial endeavors on the ISS
 - In exchange, the provider agrees to pay for a percentage of the Core Module design development and testing
 - NASA evaluates and then has the capability to accept as part of the initial proposal process or chooses not to accept based on the overall value trade
- Creates opportunity to reduce costs to the government while increasing opportunities for the Core Module provider to cultivate their own independent market

Commercial Module Acquisition Strategy

- NASA Responsibilities
 - Partner with US Industry for the development and operations of Commercial Module(s) where NASA offsets cost in exchange for a percent of use rights to module(s).
 - NASA does not want to be the majority stakeholder
 - Enable Commercial Partner to berth Comm Module to a Core Radial Port
 - Safety and compatibility focused insight and oversight of the development and operations with an appropriate requirement set
- Commercial Partner Responsibilities
 - Design, build, integrate, launch, demonstrate and operate a commercial module.
 - Demonstrate market growth and ability to provide products to a market of Government and commercial customers
 - Establish approach to achieve self-sustaining free flying commercial destination

Commercial Module Acquisition Strategy

- Commercial Partners keep their Intellectual Property and proprietary data
- Provide NASA access for verification, certification, safety, operations and maintenance insight/oversight.
- Firm Fixed price Milestones will be used to track developmental process, certification and operations
- Potential partnership arrangement:
 - NASA and Provider agree to scope, total price, milestones and NASA percent contribution
 - Milestone payments dependent on successful completion of milestone and evidence of providers expenditures
 - NASA gets a percent use/access rights to commercial module
 - Arrangement continues through sustaining and operations
 - When market demand matures, NASA will consider ordering services
- NASA is interested in lowest cash contribution
 - Similar to the Core, industry can propose other NASA contributions (e.g. NASA upmass, use of NASA crew time, use of ISS, long duration seat, etc.)

Transportation

- Transportation Services (Crew and Cargo) – while attached to ISS
 - NASA will continue to use CCP and CRS contracts
 - Transportation Services (Crew and Cargo) – while attached to Independent Core
- NASA will extend or recompete CCP and CRS contracts
 - Potential on ramp of additional vehicles based on market demand
- When market matures and NASA purchases services from a CLD, the transportation services will be reassessed